

IT Governance in Higher Education Institutions: A Systematic Literature Review

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ABSTRACT

Information Technology (IT) is a very important aspect for higher education institutions (HEI) in both teaching, research and administration. The managers of those intuitions are more and more aware that IT is a strategic tool for their institutions. On the other hand, IT Governance is getting attention from the practitioner and research side, given the need to govern IT extending the organisation's strategy and objectives into IT. IT Governance helps to set clear expectations, to gain participation, open communications, establish accountability and provide executive management oversight. Thus, it is important to consider IT Governance and the alignment with business strategy for HEI. In this article, authors present a Systematic Literature Review (SLR) on IT Governance in HEIs using a collection of scientific and non-conventional data (grey literature). The motivation that drives this literature review is the further development of an IT Governance framework for Tunisian universities. This article aims to define the situation of the IT Governance in other countries. This article aims to provide a map of the state of the art of IT governance in HEI in various countries. Results show a mixed situation of IT Governance in HEIs. Some countries have the support of top level management to introduce IT Governance in HEIs by adopting regulatory frameworks and common laws. But other countries relay in their strong culture of IT Governance. The different case studies presented in this review show that there is no consensus on the IT Governance framework or standard to use in HEI. An important number of institutions are implementing COBIT or ISO best practices. In the other hand, some counties have developed their own frameworks. Results show there is no a single way to implement an IT Governance framework designed for HEI. However, it is also true that there are two mandatory aspects that are necessary to implement in IT Governance deployments: firstly, establishing a committee structure for IT assets and secondly, enable effective communication between the IT, the business and the stakeholders.

KEYWORDS

Business Strategy, HEI, Information Technology, IT Governance

1. INTRODUCTION

Information Technology (IT) has become a very important for higher education institutions (HEI) in teaching, in research and in administration. IT is necessary in the way that provides support to university services. We can consider that IT is a strategic tool for universities. From this remark we can say that IT governance in HEI is critical due to the strategic aspect of IT. The main objective of IT

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Governance is to align business strategy with IT strategy (Dzombeta et al., 2014; Juiz & Toomey 2015). To do so, it is important that IT Governance includes strategies, policies, responsibilities, structures and processes for using IT within an HEI. The implementation of an IT Governance system in a HEI can determine who is responsible for IT strategic planning, decision making and the exploitation of IT. This way the cost in IT investment will be mastered by the introduction of project management and prioritization of investments (Lucio-Nieto et al., 2012). It is important to look in the literature to identify best practices in this domain to implement a more efficient IT Governance framework for a HEI. The motivation that drives this literature review is the development of an IT Governance framework for Tunisian universities within the context of an Erasmus + KA2 Capacity building project entitled "Information Technology governance for Tunisian Universities (IT Governance4TU)". This work aims to define the situation outside the consortium of the project. Hence, studies describing IT Governance in European countries as Spain, Germany and Norway will not be included in this review. This work aims to provide a map of the state of the art of IT governance in HEI in various countries.

After reviewing the research studies on IT Governance, a unique review study on IT Governance in higher education institution was found (Valverde-Alulema, Meza-Bolaños, & Mejia-Madrid, 2017). This work, published in Spanish, shows the interest in this subject in the scientific community. It focuses on analysing the practice of IT portfolio. Nevertheless, it does not answer to specific research questions. It does not illustrate the current state of maturity and best practices in different HEIs. The study conducted in fig work will give a map of IT Governance in higher education institutions in different countries.

The remainder of this paper is organized as follows: Section 2 describes the method used to perform the systematic literature review. The search results are presented in section 3. Finally, the findings about IT Governance in Higher Education Institutions in several countries are discussed.

2. METHOD

The state of the art on IT Governance in HEI is realized as a systematic literature review (SLR) based on the guidelines as proposed by (Kitchenham, 2007). A SLR is a means of synthesis of best quality scientific works on a specific topic or research question. The steps in the systematic literature review method are detailed below.

2.1. Research Questions

The review focuses on five research questions:

- What is the state of the university before implementing IT Governance?
- How much advanced is the implementation of IT Governance in the HEI?
- What framework/standards of IT Governance are implemented in the HEI? How the implementation process was executed?
- How IT is aligned to the university strategy?
- What best practices can be learnt from the IT Governance implementation? What tools are used for IT alignment, IT portfolio...?

2.1.1. Search Process

The search process was a manual search on specific online databases. The selected sources are shown in table 1.

The same request was used in the search engines of the different literature sources. The search query was (" Information technology governance" OR" IT governance" OR" ICT governance") AND (" higher education"). The term university was discarded from the search query since it usually appears in the affiliation section of the paper.

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